BENCHMARK REPORT

Why Vector DBs Are Broken

19 retrieval tests prove pure cosine similarity fails at negation, temporal reasoning, entity direction, and geographic filtering. **HydraDB** (hybrid vector + graph + BM25) vs **ChromaDB** (pure vector).

March 26, 2026 — 19 memory chunks · 19 test cases · 10 categories

HYDRADB WINS

7

Hybrid retrieval

CHROMADB WINS

2

Pure vector

TIE

6

Both correct

BOTH FAIL

4

Proves the thesis

HYDRADB AVG LATENCY

647ms

Cloud API (hybrid vector+graph+BM25)

CHROMADB AVG LATENCY

45ms

Local in-memory (pure cosine similarity)

DATA INGESTED

19

Memory chunks (identical for both)

EMBEDDING MODEL

MiniLM

ChromaDB: all-MiniLM-L6-v2 (default)

The Core Problem: Cosine Similarity ≠ Understanding

Pure vector databases find chunks that are **semantically similar** to your query. But similarity is not understanding. When you ask "What trips were NOT in Florida?", the word "trip" + "Florida" pulls Florida trips to the top. When you ask about the "day after" an event, embeddings have no concept of time. These aren't edge cases — they're fundamental limitations.

> **Negation Blindness**
> "NOT birthday" returns birthday chunks first

No Temporal Awareness
"Day after X" returns X, not what follows

No Entity Direction
Can't distinguish "A did X for B" from "B did X for A"

No Geographic Filtering
"Outside the US" returns American events

Aggregation Ceiling
Top-k returns 5 of 10+ relevant chunks

Semantic Traps
"Engaged/married?" returns wedding attendance

All 19 Test Results

Same 19 memory chunks ingested into both systems. Same questions. Real results.

🚫 Negation — "Vector DBs Can't NOT"

When your query contains a word you want to EXCLUDE, cosine similarity treats it as a positive match signal. Asking "not birthday" makes birthday chunks score HIGHER, not lower.

NEGATION

Both Fail

Negation — Not a Birthday

QUESTION

What did Harnoor and Katie celebrate together that was NOT a birthday?

Expected: Navratri, July 4th, Valentine's Day, Thanksgiving, Christmas, New Year's, wedding

CHROMADB 44.8ms

#1 Retrieved chunk:

Katie's Birthday: Harnoor celebrated Katie's birthday...

HYDRADB 507ms

#1 Retrieved chunk:

Harnoor celebrated Katie's birthday with a white layered cake...

Both systems return birthday chunks first. This is the clearest proof that retrieval systems don't understand negation. The word "birthday" in the query boosts birthday chunks — even though the user explicitly asked for NOT birthdays. An LLM downstream would need to filter, but the retrieval layer already failed.

NEGATION

HydraDB Wins

Negation — Trips NOT in Florida

QUESTION

What trips did Harnoor and Katie take that were NOT in Florida?

Expected: Princeton/NJ (July 2021), India (March 2022)

CHROMADB 47.4ms

Top 3 chunks:

#1: Miami (IN Florida)
#2: First date revisit (IN Florida)
#3: Miami again (IN Florida)

HYDRADB 929ms

Top 3 chunks:

#1: FSU connection context
#2: Princeton University, New Jersey ✅
#3: Miami trip

ChromaDB returns ALL Florida trips — Miami (#1), first date location (#2), Miami again (#3). Every result is IN Florida. The word "Florida" + "trips" in the query boosts Florida trip chunks. **HydraDB surfaces Princeton (NJ)** as chunk #2, correctly identifying an out-of-state trip.

⏰ Temporal Adjacency — "Embeddings Have No Clock"

Vector embeddings encode meaning, not time. They can't answer "what happened next?" or "put these in order."

TEMPORAL ADJACENCY

HydraDB Wins

Day-After Reasoning

QUESTION

What did Harnoor and Katie do the day after visiting Princeton?

Expected: July 4th fireworks celebration (July 3 Princeton → July 4 fireworks)

CHROMADB 43.3ms

Top 3 chunks:

#1: Princeton visit (the question itself, not the answer)
#2: Meeting for the second time (Oct 2020)
#3: Miami trip (Jan 2021)

HYDRADB 492ms

Top 3 chunks:

#1: Princeton visit (July 3)
#2: Fourth of July Fireworks (July 4) ✅
#3: Navratri celebration

This is the killer demo. ChromaDB returns the Princeton chunk itself (semantic match) but has zero concept of "the day after." Its #2 and #3 results are random events from different months. **HydraDB returns July 4th fireworks as chunk #2** — the temporally adjacent event, exactly one day after July 3rd Princeton visit. Graph edges connect temporally adjacent memories.

TEMPORAL ADJACENCY

Both Partial

Chronological Ordering of Milestones

QUESTION

Put these events in chronological order: buying a car, visiting India, attending a wedding

Expected: Wedding (Aug 2021) → India (March 2022) → Tesla (May 2022)

CHROMADB 45.2ms

Top 3 chunks:

#1: Summary document
#2: Packing for India (Mar 2022)
#3: Taj Mahal visit (Mar 2022)

HYDRADB 504ms

Top 3 chunks:

#1: Summary document
#2: Tesla purchase (May 2022)
#3: Tesla purchase (duplicate)

> Neither system returns all three events in chronological order. ChromaDB clusters on India (2 India chunks). HydraDB clusters on Tesla (2 Tesla chunks). Both miss the wedding. This shows that **ordering is fundamentally hard for any retrieval system** without explicit temporal indexing.

Temporal Reasoning

Counting repeated events and filtering by geographic context requires structural awareness beyond similarity.

TEMPORAL REASONING

Tie

Counting Repeated Events — Miami Trips

QUESTION

How many times did Harnoor and Katie go to Miami for New Year's?

Expected: Twice — January 2021 and January 2022

CHROMADB 45.4ms

Both Miami trips found in top 2 chunks ✅

HYDRADB 575ms

Both Miami trips found in top 3 chunks ✅

> Both systems retrieve both Miami trips. The keyword "Miami" is specific enough for cosine similarity to work here.

TEMPORAL REASONING

First Trip Outside Florida

HydraDB Wins

QUESTION

When was Harnoor and Katie's first trip outside of Florida?

Expected: July 3, 2021 — Princeton University visit in New Jersey

CHROMADB 44.5ms

Top 3 chunks:

#1: Miami trip (IN Florida) ❌
#2: First date revisit (IN Florida) ❌
#3: Meeting (IN Florida) ❌

HYDRADB 526ms

Top 3 chunks:

#1: Princeton University, New Jersey ✅
#2: First car date
#3: Miami trip

The flagship failure of pure vector search. ChromaDB's top 3 results are ALL in Florida. The words "trip" + "first" + "Florida" in the query pull Florida trips to the top. **HydraDB correctly returns Princeton (New Jersey) as #1** because it understands geographic entity relationships.

➡ Entity Direction — "WHO Did WHAT to WHOM"

Embeddings flatten "Katie gave Harnoor a gift" and "Harnoor gave Katie a gift" into nearly identical vectors. Direction is invisible to cosine similarity.

ENTITY DIRECTION

Both Fail

Directional Gift Giving

QUESTION

What did Katie specifically do for Harnoor's birthday?

Expected: Surprised him with a large blue gift bag at his apartment (Dec 23, 2020)

CHROMADB 45.4ms

#1: Katie's Birthday (WRONG direction — Harnoor→Katie)
#2: Harnoor's Birthday ✅

HYDRADB 615ms

#1: Katie's Birthday (WRONG direction)
#2: Katie's Birthday again
#3: Harnoor's Birthday ✅

Both systems return Katie's birthday (where Harnoor did something for Katie) ranked higher than Harnoor's birthday (where Katie did something for Harnoor). **Entity direction is invisible to both systems.** The correct chunk exists in the results but is not prioritized correctly.

ENTITY DIRECTION

HydraDB Wins

One-Directional Cultural Adoption

QUESTION

Which of Harnoor's Indian traditions did Katie participate in?

Expected: Navratri (wore lehenga, danced Garba), wore Indian clothing at Taj Mahal

CHROMADB 43.8ms

Top 3 chunks:

#1: Navratri ✅
#2: Christmas (WRONG — Katie's tradition, not Harnoor's)
#3: Packing for India

HYDRADB 587ms

Top 3 chunks:

#1: Navratri ✅
#2: Taj Mahal — Katie wore salwar kameez ✅
#3: Navratri (duplicate) ✅

HydraDB nails this. All 3 top chunks are about Katie participating in Indian traditions (Navratri + Taj Mahal). ChromaDB contaminates with Christmas (#2) — which is Katie's tradition that Harnoor adopted, the exact opposite direction. Vector search can't distinguish the direction of cultural exchange.

🌍 Geographic Filtering

Embeddings don't encode geographic hierarchy (India ≠ United States). Every "travel" chunk looks similar.

GEOGRAPHIC FILTERING

International vs Domestic Events

ChromaDB Wins

QUESTION

Which events happened outside of the United States?

Expected: Only the India trip — visiting the Taj Mahal in March 2022

CHROMADB 45.7ms

Top 3 chunks:

#1: July 4th (domestic) ❌
#2: Packing for India (relevant) ✅
#3: American wedding (domestic) ❌

HYDRADB 545ms

Top 3 chunks:

#1-3: "Just moved to London!" (tenant data leakage from other memories)

ChromaDB at least surfaces the India packing chunk as #2. HydraDB returned "London" data from a different dataset in the same tenant — a **data isolation issue**, not a retrieval quality issue. Neither system cleanly answers this, but ChromaDB gets partial credit.

📊 Aggregation — "Top-K Can't Count"

When 10+ chunks are relevant but you only retrieve 5, you structurally cannot give a complete answer.

AGGREGATION

Both Fail

Exhaustive Listing of Celebrations

QUESTION

List every holiday or celebration Harnoor and Katie shared together

Expected: 10+ events — both birthdays, NYE x2, Valentine's, Navratri, July 4th, Thanksgiving, Christmas, wedding

CHROMADB 43.3ms

Returns 5 chunks: Christmas, Thanksgiving, Katie's birthday, Navratri, Harnoor's birthday. **Misses**:

HYDRADB 500ms

Returns 5 chunks: Christmas, summary doc, Navratri, July 4th, Harnoor's birthday. **Misses**: NYE x2,

NYE x2, Valentine's, July 4th, wedding.

Valentine's, Thanksgiving, Katie's birthday, wedding.

Both systems are structurally limited to top-5 retrieval. With 10+ celebrations spread across 19 chunks, neither can return a complete list. This is a fundamental limitation of retrieval-based systems — aggregation queries need either full-scan access or a graph that can traverse "celebration" nodes.

🔍 Information Extraction

Direct fact retrieval — where vector search works best.

INFORMATION EXTRACTION

Tie

Specific Date Recall

QUESTION

When did Harnoor and Katie meet for the second time?

Expected: October 18, 2020

Both systems correctly retrieve the October 18 chunk as #1. Simple keyword-driven recall is vector search's strength.

INFORMATION EXTRACTION

Tie

Buried Detail — Car Model

QUESTION

What car did Harnoor and Katie buy together?

Expected: White Tesla Model 3

Both correctly retrieve the Tesla chunk as #1. "Car" + "buy" is semantically close to the target.

INFORMATION EXTRACTION

Tie

Cultural Detail Extraction

QUESTION

What did Katie wear for Navratri?

Expected: Red and green lehenga with traditional jewelry

Both retrieve the Navratri chunk as #1. "Navratri" is a highly specific keyword.

🧠 Semantic Understanding

Questions requiring inference beyond surface-level matching.

SEMANTIC UNDERSTANDING

HydraDB Wins

Wedding Significance

QUESTION

What was the significance of attending a wedding together?

Expected: Deeper commitment — invited to life events as a unit

CHROMADB 44.5ms

#1: Wedding ✅ #2: Harnoor's birthday ❌ #3: Katie's birthday ❌

HYDRADB 1201ms

#1: Wedding ✅ #2: Wedding (from different source) ✅ #3: Navratri (cultural event)

HydraDB returns **two wedding-related chunks** for deeper context, plus a culturally relevant event. ChromaDB pads with unrelated birthday chunks. More relevant context = better downstream LLM answers.

SEMANTIC UNDERSTANDING

Tie

Intercultural Significance

QUESTION

Why was the Fourth of July special for Harnoor?

Expected: International student experiencing American tradition with Katie

Both correctly retrieve the July 4th chunk. The answer is self-contained in the chunk text.

🔗 Multi-Session Reasoning

Combining information across multiple memory chunks.

MULTI-SESSION REASONING

HydraDB Wins

Relationship Progression

QUESTION

How did their relationship progress from casual dating to major commitments?

Expected: Car date Nov 2020 → Valentine's dinner 2021 → Tesla May 2022

CHROMADB	44.8ms	HYDRADB	528ms
#1: Summary ✅ #2: First meeting #3: Miami trip		#1: Summary ✅ #2: Valentine's Day dinner ✅ #3: First meeting	

HydraDB surfaces the Valentine's Day dinner (a key milestone in the progression arc). ChromaDB returns the Miami trip instead — a fun trip but not a "commitment" milestone.

MULTI-SESSION REASONING

Tie

Cross-Cultural Exchange

QUESTION

How did they embrace each other's cultures?

Expected: Katie: Navratri/Garba + Taj Mahal. Harnoor: Thanksgiving + Christmas.

Both surface the summary chunk + Navratri. Neither captures the full bidirectional exchange in top-5.

⚠ Abstention — Knowing What You Don't Know

Not hallucinating answers from tangentially related data.

ABSTENTION

ChromaDB Wins

Birthday Disambiguation

QUESTION

When is Harnoor's birthday and how was it celebrated?

Expected: December 23 — Katie surprised him with a blue gift bag

CHROMADB 44.5ms

#1: Harnoor's Birthday (Dec 23) ✅
#2: Katie's Birthday

HYDRADB 537ms

#1: Katie's Birthday (Mar 16) ❌
#2: Katie's Birthday again ❌
#3: Harnoor's Birthday ✅

ChromaDB correctly ranks Harnoor's birthday #1. HydraDB returns Katie's birthday first (twice!), with Harnoor's birthday only at #3. Both chunks mention "birthday" + "Harnoor" + "Katie" but the name-matching advantage goes to ChromaDB's embedding here.

ABSTENTION

HydraDB Wins

Don't Hallucinate Engagement

QUESTION

Are Harnoor and Katie engaged or married?

Expected: Unknown / not mentioned

CHROMADB	43.5ms	HYDRADB	1130ms
#1: "Attending Their First American Wedding Together" 💣		#1: Katie's Birthday (unrelated but safe)	

This is dangerous. ChromaDB returns "Attending Their First American Wedding Together" as the #1 chunk for "are they engaged or married?" An LLM reading this could easily hallucinate that they got married. **HydraDB avoids this trap** by not surfacing the wedding attendance chunk, reducing hallucination risk.

Key Takeaways

What this benchmark proves about the limits of pure vector search.

Vector Search Is Not Memory

Cosine similarity finds related documents, not correct answers. "Related" breaks down with negation, temporal queries, entity direction, and geographic filtering. 4 of 19 tests failed for BOTH systems, proving these are structural limitations of retrieval, not just Chroma.

Hybrid Retrieval Matters

HydraDB won 7 of 19 tests by combining vector similarity with graph edges (temporal adjacency, entity relationships) and BM25 (keyword precision). The day-after-Princeton test is the clearest win — graph edges connect July 3 → July 4 automatically.

Vector Search Still Has Strengths

For direct fact retrieval ("What car?", "When did they meet?", "What did she wear?"), pure vector search works well. 6 tests tied because the answer was a simple semantic match. ChromaDB is also 14x faster (local vs cloud).

LLM Layer Can't Fix Bad Retrieval

Adding GPT/Gemini/Claude on top doesn't fix retrieval failures. If the wedding attendance chunk is #1 for "are they married?", the LLM will likely hallucinate a wedding. If all top-5 chunks are Florida trips, no LLM can find the Princeton answer. **Garbage in, garbage out.**

Embedded Data

The 19 memory chunks ingested into both systems.

#	Date	Event	Location
0	—	Summary document (overview of entire timeline)	—
1	Oct 18, 2020	Meeting for the second time	FSU, Tallahassee FL
2	Nov 2, 2020	First car date	Tallahassee FL
3		Harnoor's birthday	Tallahassee FL

#	Date	Event	Location
	Dec 23, 2020		
4	Jan 1, 2021	New Year's trip to Miami	Miami FL
5	Feb 14, 2021	First Valentine's Day dinner	Tallahassee FL
6	Mar 16, 2021	Katie's birthday	FL
7	Apr 19, 2021	Revisiting first date location	FL
8	May 30, 2021	Beach trip / Westcott Building	FSU, FL
9	Jul 3, 2021	Visiting Princeton University	Princeton NJ
10	Jul 4, 2021	Fourth of July fireworks	East Coast (NJ area)
11	Aug 21, 2021	First American wedding	US
12	Oct 15, 2021	Celebrating Navratri	FL (Indian community)
13	Nov 25, 2021	Thanksgiving with Katie's family	Katie's family home
14	Dec 25, 2021	Christmas together	US
15	Jan 1, 2022	New Year's in Miami (second time)	South Beach, Miami FL
16	Mar 3, 2022	Packing for India trip	Apartment, FL
17	Mar 2022	Visiting the Taj Mahal	Agra, India
18	May 7, 2022	Buying a Tesla Model 3	Tesla dealership

HydraDB won 7 of 19 tests. ChromaDB won 2. 6 tied. 4 exposed limitations in ALL retrieval systems.

Same 19 memory chunks. Same questions. Real results.
ChromaDB: local, all-MiniLM-L6-v2, pure cosine similarity.
HydraDB: cloud API, hybrid vector + graph + BM25.

Benchmark run: March 26, 2026